Simplex Stream Takehome: Geometric Representations of Non-Ergodic Processes

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Abstract

Recent work has demonstrated that transformers trained via next-token prediction learn to factor their world into parts, representing concurrent, conditionally independent latent processes in orthogonal subspaces. I explore how these geometric inductive biases behave in a purely non-ergodic setting—where generative factors are mutually exclusive rather than concurrent. I trained a small decoder-only transformer on a mixture of three distinct Mess3 processes and analyzed the residual stream geometry. My pre-registered prediction favored orthogonal subspaces with explicit domain routing, but the empirical results reveal a more nuanced strategy: the model maintains *simultaneous belief trajectories under all process hypotheses*—a soft Bayesian mixture rather than hard routing. A linear probe recovers these all-process belief states with $R^2 = 0.93$ at the first layer, while process classification remains near chance, demonstrating that the model solves the non-ergodic mixture implicitly through parallel belief tracking rather than explicit process identification.

1 Introduction and Setup

I constructed a non-ergodic training dataset consisting of three distinct Mess3 processes. A single Mess3 process is a 3-state Hidden Markov Model emitting tokens from a ternary alphabet $\mathcal{X} = \{0, 1, 2\}$, parameterized by transition variables α and x, with dependent quantities $\beta = (1-\alpha)/2$ and $y = 1-2x$. The token-labeled transition matrices $T^{(0)}, T^{(1)}, T^{(2)}$ are defined as in Shai et al. (2026), Appendix C.1.1. To enforce non-ergodicity, each training sequence of length 16 was generated entirely from a single, uniformly sampled Mess3 process.

I trained a 2-layer HookedTransformer ($d_{\text{model}} = 64$, 4 heads, $d_{\text{head}} = 16$, $d_{\text{MLP}} = 256$) using standard next-token cross-entropy loss. The model was trained for 200 epochs on 50,000 samples with a batch size of 256 and learning rate 1×10^{-3}, converging to a final loss of 0.995 on a single T4 GPU.

2 Relevance to Language Models

The global training corpus of an LLM is highly non-ergodic: it contains mutually exclusive domains such as Python code, French literature, and mathematics. However, within a single context window, the generating process is typically fixed and ergodic. An LLM must therefore perform a dual-phase inference: first, rapidly identify the overarching domain from initial tokens (*macroscopic routing*), then track the specific local state within that domain (*microscopic state tracking*). A mixture of mutually exclusive Mess3 processes isolates this exact dual-phase challenge in a controlled, analytically tractable setting.

3 Pre-Registered Predictions

3.1 Mathematical Derivation of the State Space

Let $\mathcal{S}_k$ represent the set of hidden states for the k-th Mess3 process. Each process is a 3-state HMM whose predictive vector (belief state) lives in a 2-simplex spanning 2 dimensions.

In the concurrent factor setting of Shai et al., the joint latent space is the tensor product of constituent spaces. In contrast, my non-ergodic process dictates that components never co-occur. The global latent space is the disjoint union:

$$\mathcal{S}_{\text{global}} = \bigsqcup_{k=1}^{3} \mathcal{S}_k \tag{1}$$

The global belief state is a joint distribution over both the active process k and hidden state s:

$$\eta_{\text{global}} = P(\text{process } k, \text{state } s) \tag{2}$$

This spans $3 \times 3 = 9$ mutually exclusive states, so the global predictive vector lives in the 8-simplex Δ^8, requiring at most $9 - 1 = 8$ dimensions.

3.2 Alternative Geometric Hypotheses

I considered three architectures the model could adopt:

- *Flat 8-Simplex:* All 9 states represented as a flat categorical distribution with no hierarchical structure.
- *Dimensional Reuse (Superposition):* The three processes share the same 2-dimensional plane, superimposing their simplices to maximize dimensional efficiency.
- *Orthogonal Subspaces:* Distinct, mutually orthogonal 2-simplices per process, separated by a global routing vector.

3.3 Geometric Intuition and Dynamics

My intuition favored orthogonal subspaces: 2 dimensions for a macroscopic categorical distribution over processes, and 6 dimensions for three orthogonal 2-simplices. I expected:

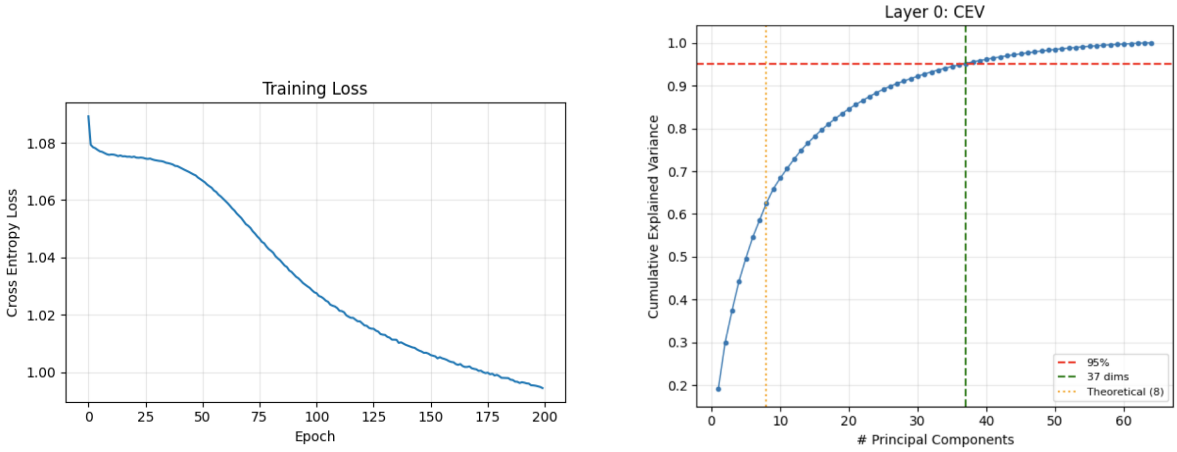
- *Across context position:* Activations collapse near a global center of mass at $l = 0$ and progressively separate into process-specific clusters.
- *Across layers:* Early layers solve routing; later layers refine belief geometry.

4 Residual Stream Geometry Analysis

To test these predictions, I extracted residual stream activations at the final sequence position across a 2,000-sample validation set.

The loss curve (Figure 1a) confirms the model learns non-trivial Markov structure. The CEV analysis (Figure 1b) shows the final layer requires 28 dimensions for 95% variance, still well above the theoretical 8-dimensional bound. This indicates the model has not yet fully compressed to the optimal geometry.

PCA projections (Figure 2) reveal that Process 3 ($x = 0.50$, $y = 0$) separates macroscopically in PC1–PC2, while Processes 1 and 2—which share similar dynamics ($x \in \{0.11, 0.15\}$)—remain largely overlapping. The microscopic geometry (PC3–PC4) does not yet show crisp 2-simplex structure.



(a) Cross-entropy loss over 200 epochs

(b) Cumulative Explained Variance (Layer 1)

Figure 1: Training dynamics and effective dimensionality. The model converges to a loss of 0.995, well below the uniform entropy $\log 3 \approx 1.099$. The final layer requires 28 dimensions for 95% variance—above the theoretical 8-dimensional bound, but reduced from 37 at Layer 0.

4.1 Context Position Evolution

As predicted, early positions show a collapsed geometry near a global center of mass (Figure 3, $l = 1$). As context accumulates, the representations progressively separate into process-specific clusters, confirming that the model actively separates generative domains as evidence accrues.

4.2 Linear Probes: Belief States and Process Identification

To determine what the model *actually* represents, I constructed probe targets that reflect what a Bayes-optimal predictor must maintain. For a non-ergodic mixture, the optimal next-token distribution is:

$$P(x_{\ell+1} \mid x_{1:\ell}) = \sum_{k=1}^{3} P(k \mid x_{1:\ell}) \cdot P(x_{\ell+1} \mid x_{1:\ell}, \text{process } k) \tag{3}$$

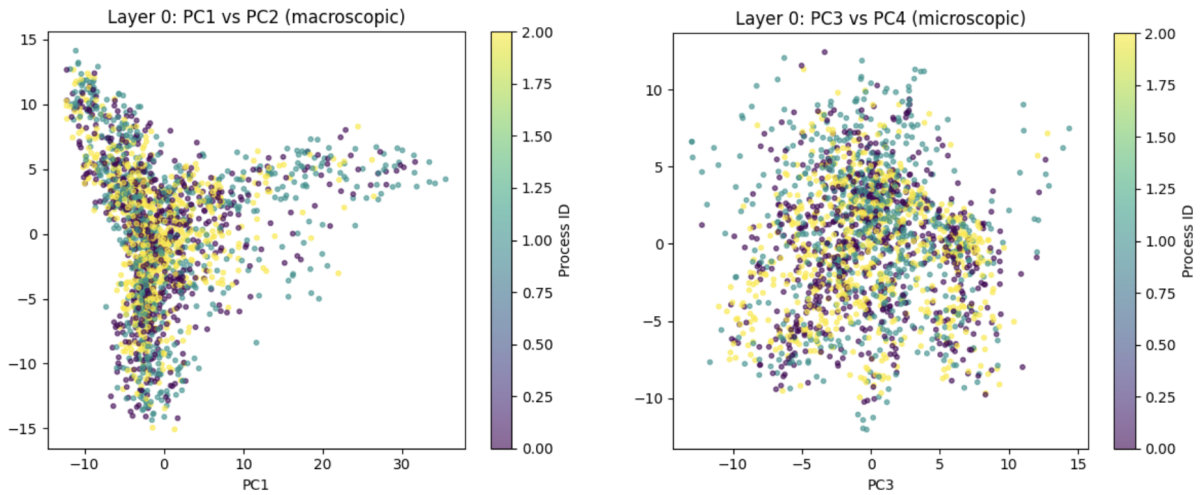
where $P(x_{\ell+1} \mid x_{1:\ell}, k)$ depends on $\eta_k(x_{1:\ell})$—the belief state obtained by running the HMM forward algorithm under process k's transition dynamics on the observed tokens, *regardless of which process actually generated the data.* I computed $\eta_k(x_{1:\ell})$ for all three processes on every validation sequence, yielding a 9-dimensional target $[\eta_1, \eta_2, \eta_3]$. I also computed the Bayesian posterior $P(k \mid x_{1:\ell})$ via cumulative log-likelihoods with a uniform prior.

Probe	**Layer 0**	**Layer 1**
Belief State R^2 (all-process)	**0.929**	0.633
Process Classification Accuracy	0.393	0.460
Random baseline R^2	-0.047	
Chance classification	0.333	

Table 1: Per-layer probing results. Layer 0 achieves remarkably high belief R^2 while process classification remains near chance at both layers.

The results (Table 1) overturn my pre-registered prediction in a revealing way:

1. **Belief tracking is strong; routing is weak.** The linear probe recovers all-process belief states with $R^2 = 0.929$ at Layer 0 and 0.633 at Layer 1 (per-process breakdown:



(a) Macroscopic geometry (PC1 vs PC2)

(b) Microscopic geometry (PC3 vs PC4)

Figure 2: PCA projections of Layer 1 residual stream at the final sequence position, colored by generative process ID.

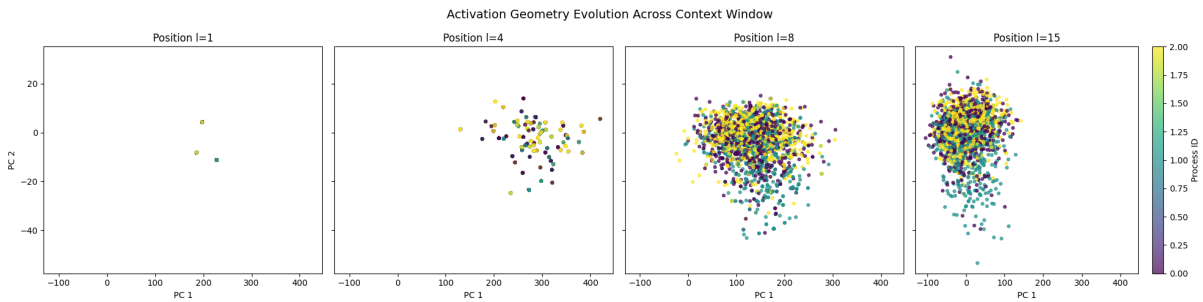


Figure 3: Activation geometry evolution across the context window ($l = 1, 4, 8, 15$), projected onto the Layer 1 PCA basis.

$R^2_{\text{P1}} = 0.672$, $R^2_{\text{P2}} = 0.655$, $R^2_{\text{P3}} = 0.571$). Meanwhile, process classification barely exceeds chance (0.46 vs. 0.33). The model tracks belief states under all three process hypotheses simultaneously without explicitly identifying the active process.

2. **Belief readability peaks at Layer 0, not Layer 1.** This is the opposite of the predicted "early routing, late refinement" pattern. The first transformer block builds a representation optimized for belief tracking; the second block transforms it for logit computation, compressing the belief geometry into a less linearly readable form.

3. **The Bayesian posterior explains the weak routing.** The mean Bayes-optimal posterior confidence is only 0.586—even with perfect information, process identification from 15 tokens averages only ∼59% confidence. The three Mess3 processes produce heavily overlapping token distributions, making hard routing fundamentally unreliable. The model's strategy of maintaining parallel beliefs is the rational response to this ambiguity.

This reveals a fundamentally different solution from the one I predicted. Rather than the "orthogonal subspaces" hypothesis, the model adopts a strategy closer to *dimensional reuse*—maintaining belief trajectories for all processes in a shared representational space, weighted implicitly by the Bayesian posterior. The high subspace overlaps in the next section corroborate this interpretation.

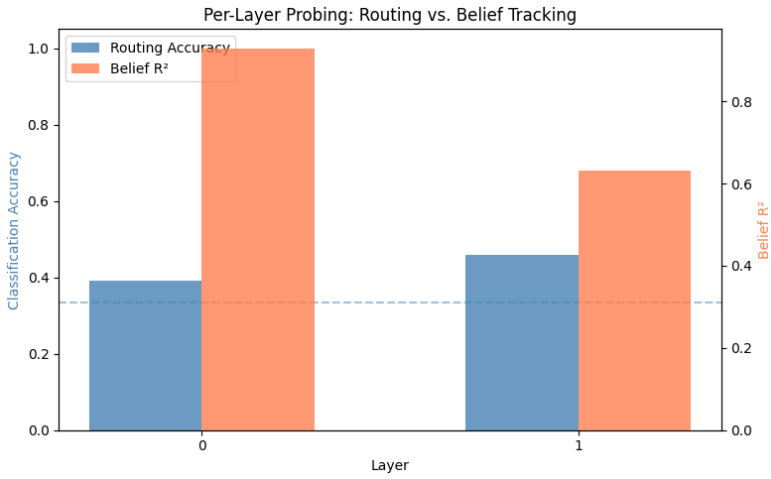
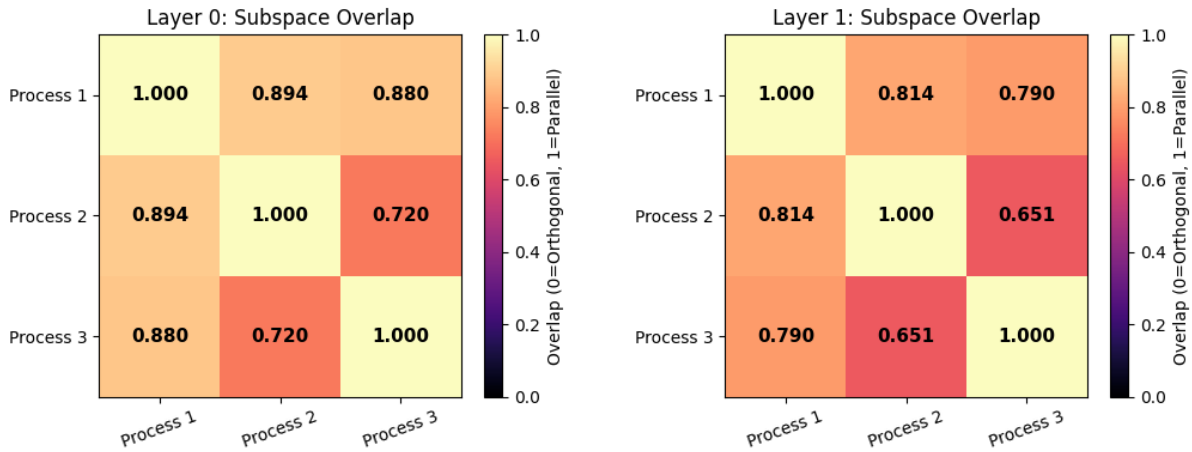


Figure 4: Per-layer probing: belief tracking peaks at Layer 0 ($R^2 = 0.93$) while process classification remains near chance at both layers. Dashed line indicates chance (1/3).

5 Additional Analysis: Layer-Wise Subspace Overlap

5.1 Motivation

To quantify how the model organizes process-specific information geometrically, I computed the pairwise subspace overlap between the top-2 PCA subspaces of activations grouped by process ID. Following the metric in Shai et al. (Appendix H.2.2), overlap ranges from 0 (perfectly orthogonal) to 1 (fully overlapping).



(a) Layer 0 Subspace Overlap

(b) Layer 1 Subspace Overlap

Figure 5: Pairwise subspace overlap between ergodic components at each layer.

5.2 Results

The overlap analysis (Figure 5) reinforces the soft mixture interpretation:

- All pairwise overlaps remain high (0.65–0.89), consistent with processes sharing representational space rather than being orthogonalized into distinct subspaces.
- The isolation is *asymmetric*: P2–P3 achieves the lowest overlap (0.72 at Layer 0, 0.65 at Layer 1), while P1–P2 remains most overlapping (0.89 → 0.81). This aligns with the

generative parameters: P1 ($\alpha = 0.60, x = 0.15$) and P2 ($\alpha = 0.79, x = 0.11$) share similar transition dynamics, while P3 ($x = 0.50$, $y = 0$) has fundamentally different entropy.

- All overlaps decrease from Layer 0 to Layer 1, indicating a nascent trend toward orthogonalization that has not yet completed—consistent with an extended geometric crystallization phase.

6 Conclusion and Future Work

My pre-registered prediction—orthogonal subspaces with explicit routing—was wrong in an informative way. The model instead discovered a *soft Bayesian mixture* strategy: maintaining belief trajectories under all process hypotheses simultaneously, without explicitly identifying the active process. This is the rational response to the inherent ambiguity of the non-ergodic mixture (mean Bayesian posterior confidence of only 59%).

This finding has a natural analogue in language models: when early context is ambiguous between domains, the model may maintain parallel hypotheses rather than committing to a hard routing decision. The orthogonal factored representation that Shai et al. observe for concurrent, conditionally independent factors does not straightforwardly emerge for mutually exclusive factors—likely because the conditional independence property (Definition 2.1 in the paper) does not hold for a non-ergodic mixture. Observing a token simultaneously updates both the process posterior and within-process beliefs in a coupled manner, preventing clean factorization.

Given more time, I would: (1) train substantially longer to test whether the 28-dimensional representation eventually compresses toward 8 and whether orthogonal subspaces emerge under sustained gradient pressure; (2) vary the number and distinguishability of the Mess3 components to map the transition between soft mixture and hard routing strategies; and (3) perform attention pattern analysis to determine whether specific heads specialize in routing vs. belief tracking within the shallow architecture.